

**B.Tech. DEGREE EXAMINATION, NOVEMBER 2011**  
**DYNAMICS OF MACHINERY**

**PART A - (10 x 2 = 20 marks)**

- 1) State D'Alembert's principle.
- 2) What is meant by piston effort and crank effort?
- 3) The piston of a steam engine moves with SHM. The crank rotates at 120 rpm with a stroke of 2m. Find the velocity and acceleration of the piston, when it is at a distance of 0.75 m from the centre.
- 4) Define forced vibrations and damped vibrations.
- 5) Define torsionally equivalent shaft.
- 6) Derive an expression for natural frequency for two and three rotor systems.
- 7) What is meant by hunting?
- 8) Define gyroscopic couple.
- 9) State the effects of partial balancing in locomotives.
- 10) What are the types of balancing machines?

**PART B - (5 x 11 = 55 marks)**

**UNIT I**

- 11) In a vertical double-acting steam engine, the connecting rod is 4.5 times the crank. The weight of the reciprocating parts is 120 kg and the stroke of the piston is 440 mm. The engine runs at 250 rpm. If the net load on the piston due to steam pressure is 25 kN when the crank has turned through an angle of  $120^\circ$  from the top dead centre, determine the
  - a) Thrust in the connecting rod
  - b) Pressure in the connecting rod
  - c) Tangential force on the crank pin
  - d) Thrust on the bearings
  - e) Turning moment on the crankshaft.

**[or]**

- 12) A three cylinder single - acting engine has its cranks at  $120^\circ$ . The turning moment diagram for each cycle is a triangle for the power stroke with a maximum torque of 60 Nm at  $60^\circ$  after the dead centre of the corresponding crank. There is no torque on the return stroke. The engine runs at 400 rpm. Determine:

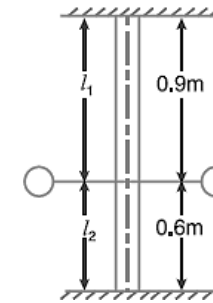
- a) Power developed
- b) Coefficient of fluctuation of speed if the mass of the flywheel is 10 kg and radius of gyration is 88 mm.
- c) Coefficient of fluctuation of energy
- d) Maximum angular acceleration of flywheel.

**UNIT II**

- 13) A uniform slender rod 1.2 m long is fitted with a transverse pair of knife-edges, so that it can swing in a vertical plane as a compound pendulum. The position of the knife edges is variable. Find the time of swing of the rod, if (a) the knife edges are 50 mm from one end of the rod, and (b) the knife edges are so placed that the time of swing is minimum. In case (a) find also the maximum angular velocity and the maximum angular acceleration of the rod if it swings through  $3^\circ$  on either side of the vertical.

**[or]**

- 14) A flywheel is mounted on a vertical shaft as shown in fig. The both ends of the shaft are fixed and its diameter is 50 mm. The flywheel has a mass of 500 kg. Find the natural frequencies of longitudinal and transverse vibrations. Take  $E = 200 \text{ GN/m}^2$ .



**UNIT III**

- 15) A steel shaft ABCD 1.5 m long has flywheel at its ends A and D. The mass of the flywheel A is 600 kg and has a radius of gyration of 0.6 m. The mass of the flywheel D is 800 kg and has a radius of gyration of 0.9 m. The connecting shaft has a diameter of 50 mm for the portion AB which is 0.4 m long; and has a diameter of 60 mm for the portion BC which is 0.5 m long; and has a diameter of d mm for the portion CD which is 0.6 m long.

Determine :

- a) the diameter 'd' of the portion CD so that the mode of the torsional vibration of the system will be at the centre of the length of the BC,
- b) the natural frequency of the torsional vibrations. The modulus of rigidity for the shaft material is  $80 \text{ GN/m}^2$ .

**[or]**

- 16) An electric motor rotating at 1500 rpm drives centrifugal pump at 500 rpm through a single stage reduction gearing. The moments of inertia the electric motor and the pump impeller are  $400 \text{ kg-m}^2$  and  $1400 \text{ kg-m}^2$  respectively. The motor shaft is 45 mm in diameter and 180 mm long. The pump shaft is 90 mm in diameter and 450 mm long. Determine the frequency of torsional, oscillation of the system, neglecting the inertia of the gears. The modulus of rigidity for the shaft material is  $84 \text{ GN/m}^2$ .

#### UNIT IV

- 17) For a spring controlled Hartnell type governor, following data are provided :

Mass of the governor ball = 1.8 kg,

Length of the vertical arm of bell crank lever = 87.5 mm and

Length of other arm of bell crank lever = 100 mm.

The speed corresponding to radii of rotations of 120 mm and 130 mm are 296 and 304 rpm respectively.

Determine :

- a) The initial compression of the central spring and
- b) The spring constant.

**[or]**

- 18) The propeller of aero weighs 550 N and has radius of gyration of 0.9 m. The propeller shaft rotates at 1900 rpm, clockwise as viewed from tail end. The plane turns left, making a U turn i.e. through  $180^\circ$ , of 125 m radius, at a speed of 330 kmph, determine the gyroscopic couple and its effect on the aircraft. Also find the reactions on bearings if the distance between two bearings of the propeller is 0.8m.

#### UNIT V

- 19) A two cylinder uncoupled locomotive has inside cylinders 0.6 m apart. The radius of each crank is 300 mm and is at right angles. The revolving mass per cylinder is 250 kg and the reciprocating mass per cylinder is 300 kg. The whole of the revolving and two-third of the reciprocating masses are to be balanced and the balanced masses are to be placed, in the planes of rotation

of the driving wheels a radius of 0.8 m. The driving wheel is 2 m in diameter and 1.5 m apart. If the speed of the engine is 80 km/hr, determine

- a) Hammer blow,
- b) Maximum variation in tractive force, and
- c) Maximum swaying couple.

**[or]**

- 20) A V-twin engine has the cylinder axes at right angles and the connecting rods operate a common crank. The reciprocating mass per cylinder 10 kg and the crank radius is 70 mm. The length of the connecting rod is 350 mm. Show that the engine may be balanced for primary forces by means of revolving balance mass. If the engine speed is 500 rpm, What is the value of maximum resultant secondary force?

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**B.Tech. DEGREE EXAMINATION, APRIL 2013**  
**DYNAMICS OF MACHINERY**

**PART A - (10 x 2 = 20 marks)**

- 1) State conditions for equilibrium of two force members.
- 2) Define piston effort.
- 3) What are the methods are used to find natural frequency of transverse vibration?
- 4) Give a governing equation for single degree free damped longitudinal vibration.
- 5) Explain whirling speed.
- 6) What is amplitude reduction factor?
- 7) Define hunting and isochronous.
- 8) Differentiate steering and pitching of a Naval ship.
- 9) State conditions for static and dynamic balancing.
- 10) Why complete balancing is not possible in reciprocating engine?

**PART B – (5 x 11 = 55 marks)**

**UNIT I**

- 11) A single cylinder vertical engine has a bore of 300 mm and a stroke of 400 mm. The connecting rod is 1 m long. The mass of the reciprocating parts is 140 kg. On the expansion stroke with the crank at  $30^\circ$  from the top dead centre the gas pressure is 0.7 MPa. If the engine runs at 250 r.p.m., determine:
  - a) net force acting on the piston
  - b) resultant load on the gudgeon pin,
  - c) thrust on the cylinder walls and
  - d) the speed above which, other thing remaining same, the gudgeon pin load would be reversed in direction.

**[or]**

- 12) The torque exerted on the crank shaft of a two stroke engine is given by the equation  $T(Nm) = 14500 + 2300 \sin \theta - 1900 \cos 2\theta$ , where  $\theta$  is the crank angle displacement from the inner dead centre. Assuming the resisting torque to be constant. Determine :
  - a) the power developed when the speed is 150 r.p.m.,
  - b) the moment of inertia of the flywheel if the total fluctuation of speed is not to exceed 1% of the mean speed.

**UNIT II**

- 13) The mass of a single degree damped vibrating system is 7.5 kg and makes 24 free oscillations in 14 seconds when disturbed from its equilibrium position. The amplitude of vibration reduces to 0.25 of its initial value after five oscillations. Determine:
  - a) stiffness of the spring,
  - b) logarithmic decrement, and
  - c) damping factor, i.e. the ratio of the system damping to critical damping.

**[or]**

- 14) The mass of an electric motor is 120 kg and it runs at 1500 r.p.m. The armature mass is 35kg and its C.G. lies 0.5 mm from the axis of rotation. The motor is mounted on five springs of negligible damping so that the force transmitted is one-eleventh of the impressed force. Assume that the mass of the motor is equally distributed among the five springs. Determine :
  - a) stiffness of each spring;
  - b) dynamic force transmitted to the base at the operating speed; and
  - c) natural frequency of the system.

**UNIT III**

- 15) A hollow shaft 1.8 m long is supported in flexible bearings at the ends. It carries two wheel each of 50 kg mass, one at the centre of the shaft and the other at 450 mm from the centre. The external and internal diameters of the shaft are 80 mm and 50 mm respectively. Determine the whirling speed of the shaft. The density of the material is  $7,500 \text{ kg/m}^3$  and the modulus of elasticity  $210 \text{ GN/m}^2$ .

**[or]**

- 16) A steel shaft 1.5 m long is 95 mm in diameter for the first 0.6 m of its length, 60 mm in diameter for the next 0.5 m of the length and 50 mm in diameter for the remaining 0.4 m of its length. The shaft carries two flywheels at two ends, the first having the mass of 900 kg and 0.85 m radius of gyration located at the 95 mm diameter end and the second having a mass of 700 kg and 0.55 m radius of gyration located at the other end. Determine the location of the node frequency of the free torsional vibration of the system. The modulus of rigidity of shaft material may be taken as  $80 \text{ GN/m}^2$ .

#### UNIT IV

- 17) A Hartnell governor having a central sleeve spring and two right-angled bell crank levers moves between 290 r.p.m. and 310 r.p.m. for a sleeve lift of 15mm. The sleeve arms and the ball arms are 80 mm and 120 mm respectively. The levers are pivoted at 120 mm from the governor axis and mass of each ball is 2.5 kg. The ball arms are parallel to the governor axis at the lowest equilibrium speed. Determine:
- loads on the spring at the lowest and the highest equilibrium speeds, and
  - stiffness of the spring.

[or]

- 18) The turbine rotor of a ship has a mass of 8 tonnes and a radius of gyration 0.6 m. It rotates at 1800 r.p.m. clockwise, when looking from the stern. Determine the gyroscopic couple, if the ship travels at 100 km/hr and steer to the left in a curve of 75 m radius.

#### UNIT V

- 19) A, B, C and D are four masses carried by a rotating shaft radii 100, 125, 200 and 150 mm respectively. The planes in which masses revolve are spaced 600 mm apart and the mass of B, C, and D are 10 kg, 5 kg and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance.

[or]

- 20) The cranks of a two cylinder uncoupled inside cylinder locomotive are at right angles are 300mm long. The distance between the centre lines of the cylinder is 650 mm. The wheel centre lines are 1.6 m apart. The reciprocating mass per cylinder is 300 kg. The driving wheel diameter is 1.8 m. if the hammer blow is not to exceed 45 kN at 100 km/hr, determine:
- The fraction of the reciprocating masses to be balanced,
  - the variation tractive force,
  - the maximum swaying couple.

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**B.Tech. DEGREE EXAMINATION, NOVEMBER 2013  
DYNAMICS OF MACHINERY**

**PART A - (10 x 2 = 20 marks)**

- 1) State D'Alembert's principle.
- 2) What is the function of flywheel in punching press?
- 3) Obtain the natural frequency of vibration for the system when two springs are connected in series. Take  $s_1 = 2000 \text{ N/m}$ ,  $s_2 = 1500 \text{ N/m}$  and  $m = 10 \text{ kg}$ .
- 4) Define critical damping and give an application of critical damping.
- 5) Differentiate longitudinal and transverse vibration.
- 6) What is the meaning of equivalent shaft?
- 7) How governor differs from flywheel?
- 8) What is gyroscopic torque?
- 9) Why balancing is required in high speed engine?
- 10) Define shaking force and hammer blow.

**PART B - (5 x 11 = 55 marks)**

**UNIT I**

- 11) During a trial on steam engine, it is found that the acceleration of the piston is  $36 \text{ m/s}^2$  when the crank has moved  $30^\circ$  from the inner dead centre position. The net effective steam pressure on the piston is  $0.5 \text{ N/m}^2$  and the frictional resistance is equivalent to a force of  $600 \text{ N}$ . The diameter of the piston is  $300 \text{ mm}$  and the mass of the reciprocating parts is  $180 \text{ kg}$ . If the length of the crank is  $300 \text{ mm}$  and the ratio of the connecting rod length to the crank length is  $4.5$ , find : (a) Reaction on the guide bars, (b) Thrust on the crank shaft bearings, and (c) Turning moment on the crank shaft.

**[or]**

- 12) The turning moment diagram for a multi cylinder engine has been drawn to a scale  $1 \text{ mm} = 600 \text{ N-m}$  vertically and  $1 \text{ mm} = 3^\circ$  horizontally. The intercepted areas between the output torque curve and the mean resistance line, taken in order from one end, are as follows :  $+52$ ,  $-124$ ,  $+92$ ,  $-140$ ,  $+85$ ,  $-72$  and  $+107 \text{ m}^2$ , when the engine is running at a speed of  $600 \text{ r.p.m.}$  If the total fluctuation of speed is not to exceed  $\pm 1.5\%$  of the mean, find the necessary mass of the flywheel of radius  $0.5 \text{ m}$ .

**UNIT II**

- 13) The mass of a single degree damped vibrating system is  $7.5 \text{ kg}$  and makes  $24$  free oscillations in  $14$  seconds when disturbed from its equilibrium position. The amplitude of vibration reduces to  $0.25$  of its initial value after five oscillations. Determine : (a) stiffness of the spring, (b) logarithmic decrement, and (c) damping factor. i.e. the ratio of the system damping to critical damping.

**[or]**

- 14) The mass of an electric motor is  $120 \text{ kg}$  and it runs at  $1500 \text{ r.p.m.}$  The armature mass is  $35 \text{ kg}$  and its C.G. lies  $0.5 \text{ mm}$  from the axis of rotation. The motor is mounted on five springs of negligible damping so that the force transmitted is one-eleventh of the impressed force. Assume that the mass of the motor is equally distributed among the five springs. Determine: (a) stiffness of each spring; (b) dynamic force transmitted to the base at the operating speed; and (c) natural frequency of the system.

**UNIT III**

- 15) A shaft is simply supported at its ends and is of  $40 \text{ mm}$  in diameter and  $2.5 \text{ m}$  in length. The shaft carries three point loads masses  $30 \text{ kg}$ ,  $70 \text{ kg}$  and  $45 \text{ kg}$  at  $0.5 \text{ m}$ ,  $1 \text{ m}$  and  $1.7 \text{ m}$  respectively from the left support. The weight of the shaft per meter length is given as  $73.575$ . The Young's modulus for the material of the shaft is  $200 \text{ GN/m}^2$ . Find the critical speed of the shaft.

**[or]**

- 16) A shaft of length  $1.25 \text{ m}$  is  $75 \text{ mm}$  in diameter for the first  $275 \text{ mm}$  of its length,  $125 \text{ mm}$  in diameter for the next  $500 \text{ mm}$  length,  $87.5 \text{ mm}$  in diameter for the next  $375 \text{ mm}$  length and  $175 \text{ mm}$  in diameter for the remaining  $100 \text{ mm}$  of its length. The shaft carries two rotors at two ends. The mass moment of inertia of the first rotor is  $75 \text{ kg m}^2$  whereas of the second rotor is  $\text{kg m}^2$ . Find the frequency of natural torsional vibrations of the system. The modulus of the rigidity of shaft material is  $80 \text{ GN/m}^2$ .

**UNIT IV**

- 17) The following particulars refer to a governor with open arms :  
Length of all arms =  $200 \text{ mm}$ ;  
Distance of pivot of arm from the axis of rotation =  $40 \text{ mm}$ ;  
Length of extension of lower arms to which each ball is attached =  $100 \text{ mm}$ ;  
Mass of each ball =  $6 \text{ kg}$  and mass of the central load =  $150 \text{ kg}$ .

If the radius of rotation of the balls is 180 mm when arms are inclined at an angle of  $40^\circ$  to the axis of the rotation, find the equilibrium speed of the above configuration.

**[or]**

- 18)** An aeroplane makes a complete half circle 50 metres radius, towards left, when flying 200 km per hr. The rotary engine and propeller of the plane has a mass of 400 kg and radius of gyration of 0.3 m. The engine rotates at 2400 r.p.m. clockwise when viewed from the rear. Find the gyroscopic couple on the aircraft and state its effect on it.

## **UNIT V**

- 19)** A shaft carries 5 masses  $m_1$ ,  $m_2$ ,  $m_3$ ,  $m_4$  and  $m_5$  which revolve at same radius in planes which are equal distance from one another. The magnitude of the masses in plane, 1, 3, 4 are 50 kg, 40 kg and 30 kg respectively. The position of masses 3 and 4 with respect to  $m_1$  are  $60^\circ$  and  $120^\circ$  respectively. Determine the weight 2 and 5 and the position with respect to plane in order to out the shaft in complete rotory balance.

**[or]**

- 20)** A four cylinder vertical engine has cranks 300 mm long. The planes of the rotation of the first, third and fourth cranks are 750 mm, 1050 mm and 1650 mm respectively from that of the second crank and their reciprocating masses are 150 kg, 400 kg and 250 kg respectively. Find the mass of the reciprocating parts for the second cylinder and relative angular positions of the cranks in order that the engine may be in complete primary balance.

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**B.Tech. DEGREE EXAMINATION, APRIL/MAY 2014**  
**DYNAMICS OF MACHINERY**

**PART A - (10 x 2 = 20 marks)**

- 1) State conditions for equilibrium of two force members.
- 2) Define crank effort.
- 3) Define damping and viscous damping.
- 4) What do you understand by vibration isolation?
- 5) Define critical speed.
- 6) State natural frequency of the torsional vibration of a simple system.
- 7) Define hunting and isochronous.
- 8) What is controlling force in governor?
- 9) State conditions for dynamic balancing.
- 10) Why complete balancing is not possible in reciprocating engine?

**PART B — (5 x 11 = 55 marks)**

**UNIT I**

- 11) The length of the connecting rod of a vertical double acting steam engine is 1.5 m. The diameter of the cylinder is 400 mm and stroke of the engine is 600 mm. the crank is rotating at 200 r.p.m. in the clockwise direction. The crank has turned through  $40^\circ$  from the top dead centre and piston is moving downwards. The steam pressure above the piston is  $0.6 \text{ N/mm}^2$  and below is  $0.05 \text{ N/mm}^2$ . The mass of the reciprocating parts is 200 kg. The diameter of piston rod is given as 50 mm. Find the thrust on guide bars and crank shaft bearing and also turning moment on crank shaft.

**[or]**

- 12) The turning moment diagram for a multi-cylinder engine has been drawn to a scale of 1 mm 325 Nm vertically and 1mm =  $3^\circ$  horizontally, The areas above and below the mean torque line are -26, +378, -256, +306, -302, +244, -380, +261 and -225 mm<sup>2</sup>. The engine is running at a mean speed of 600 r.p.m. The total fluctuation of speed is not to exceed  $\pm 1.8\%$  of the mean speed. If the radius of flywheel is 0.7 m find the mass of the flywheel.

**UNIT II**

- 13) In a single degree damped vibrating system, a suspended mass of 3.75 kg makes 12 oscillations in 7 second when disturbed from its equilibrium

position. The amplitude of vibration reduces to 0.33 of its initial value after four oscillations. Determine:

- a) stiffness of the spring
- b) logarithmic decrement
- c) damping factor and
- d) the damping coefficient.

**[or]**

- 14) A harmonic exciting force of 25 N is acting on a machine part, which is having a mass 2 kg and is vibrating in a viscous medium. The exciting force causes a resonant amplitude of 12.5 mm with a period of 0.20 seconds. Determine damping co- efficient. If the system is excited by a harmonic force of frequency 4 Hz, find the increase in amplitude of forced vibration when damper is removed.

**UNIT III**

- 15) A shaft 1.5 m long, supported in flexible bearings at the ends carries two wheels each of 50kg mass. One wheel is situated at the centre of the shaft and the other at a distance of 375 mm from the centre towards left. The shaft is hollow of external diameter 75 mm and internal diameter 40 mm. The density of the shaft material is  $7700 \text{ kg/m}^3$  and its modulus of elasticity is  $200 \text{ GN/mm}^2$ . Find the lowest whirling speed of the shaft, taking into account the mass of the shaft.

**[or]**

- 16) A motor drives a centrifugal pump through gearing, the pump speed being one-third that of the motor. The shaft from the motor to the pinion is 60 mm diameter and 300 mm long. The moment of inertia of the motor is  $400 \text{ kg-m}^2$ . The impeller shaft is 100 mm diameter and 600 mm long. The moment of inertia of the impeller is  $1500 \text{ kg-m}^2$ . Neglecting the inertia of the gears and shaft, determine the frequency of torsional vibration of the system. The modulus of rigidity of the material of the shaft material is  $80 \text{ GN/m}^2$ .

**UNIT IV**

- 17) The upper and lower arms of porter governor are of 200 mm and 250 mm respectively and pivoted on the axis of rotation. The control load is 160 N and weight of each ball is 20 N and friction of the sleeve together with the resistance of the operating gear is equal to a load of 25 N at the sleeve. If the limiting inclinations of the upper arms to the vertical arc  $32^\circ$  and  $42^\circ$ , find taking friction into account, range of speed of the governor.

**[or]**

- 18)** The governor of the Hartnell type has equal balls of mass 3 kg, set initially at a radius of 200 mm. The arms of the bell-crank lever are 110 mm vertically and 150 mm horizontally. Find
- the initial compressive force on the spring, if the speed for an initial ball radius of 200 mm is 240 r.p.m. and
  - the stiffness of the spring required to permit a sleeve movement of 4 mm on a fluctuation of 7.5% in the engine speed.

## **UNIT V**

- 19)** A, B, C and D are four masses carried by a rotating shaft radii 100, 125, 200 and 150 mm respectively. The planes in which masses revolve are spaced 600mm apart and the mass of B, C, and D are 10 kg, 5 kg and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance.

**[or]**

- 20)** An inside cylinder locomotive has its cylinder centre lines 0.7 m apart and has a stroke of 0.6 m. The rotating masses per cylinder are equivalent to 150 kg at the crank pin, and the reciprocating masses per cylinder to 180 kg. The wheel centre lines are 1.5 m apart. The cranks are at right angles. The whole of the rotating and  $\frac{2}{3}$  of the reciprocating masses are to be balanced by masses placed at a radius of 0.6 m. Find the magnitude and direction of the balancing masses. Find the fluctuation in rail pressure under one wheel, variation of tractive effort and the magnitude of swaying couple at a crank speed of 300 r.p.m.

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**B.Tech. DEGREE EXAMINATION, NOVEMBER 2014**  
**DYNAMICS OF MACHINERY**

**PART A - (10 x 2 = 20 marks)**

- 1) Explain the significance of inertia force analysis.
- 2) Single cylinder engine needs heavier size flywheel. Justify.
- 3) Define Transmissibility.
- 4) Write short note on degrees of freedom.
- 5) Write the expression for natural frequency of vibration using Rayleigh's method.
- 6) What do you understand by torsionally equivalent shaft?
- 7) What is the function of the Governor?
- 8) Explain the gyroscopic effects on aircrafts.
- 9) Why is partial balancing done for reciprocating masses?
- 10) Why rotating masses are to be dynamically balanced?

**PART B - (5 x 11 = 55 marks)**

**UNIT I**

- 11) The dimension of a four-link mechanism are  $AB = 500$  mm,  $BC = 660$  mm,  $CD = 560$  mm, and  $AD = 1000$  mm the link AB has an angular velocity of  $10.5$  rad/s counter clockwise and an angular retardation of  $26$  rad/s<sup>2</sup> at the instant when it makes an angle of  $60^\circ$  with AD, the fixed link. The mass of the links BC and CD is  $4.2$  kg/m length. The link AB has a mass of  $3.54$  kg, the center of which lies at  $200$  mm from A and a moment of inertia of  $88,500$  kg mm<sup>2</sup> neglecting the gravity and friction effects, determine the instantaneous value of the drive torque required to be applied on AB to overcome the inertia forces.

**[or]**

- 12) A single cylinder, single acting, four stroke cycle gas engine develops  $20$  kw at  $200$  rpm. The work done by the gases during the expansion stroke is  $3$  times the work done on the gases during compression stroke. The work done on the suction and exhaust strokes may be neglected. If the flywheel has a mass of  $1000$  kg and has a radius of gyration of  $0.6$  m, find the cycle fluctuation of energy and coefficient of fluctuation of speed.

**UNIT II**

- 13) A mass suspended from a helical spring vibrates in a viscous fluid medium whose resistance varies directly with the speed. It is observed that the frequency of damped vibration is  $90$  per minute and that the amplitude decreases to  $20\%$  of its initial value in one complete vibration. Find the frequency of the free undamped vibration of the system.

**[or]**

- 14) A harmonic exciting force of  $25$  N is acting on a machine part, which is having a mass of  $2$  kg and is vibrating in a viscous medium. This exciting force causes a resonant amplitude of  $12.5$  mm with a period of  $0.20$  seconds. Determine the damping co-efficient. If the system is excited by a harmonic force of frequency  $4$  Hz, find the increase in amplitude of forced vibration when damper is removed.

**UNIT III**

- 15) A shaft is simply supported at the ends and is of  $20$  mm in diameter and  $600$  mm in length. The shaft carries a load of  $19.62$  N at its centre. The weight of shaft per metre length is  $248.2$  N. Find the critical speed of the shaft. Take Young's modulus =  $200$  GN/m<sup>2</sup>.

**[or]**

- 16) A shaft of length  $1.5$  m is  $0.095$  m in diameter for the first  $0.6$  m of its length,  $0.06$  m in diameter for the next  $0.5$  m of the length and  $0.05$  m in diameter for the remaining  $0.4$  m of its length. The shaft carries two rotors at two ends. The mass moment of inertia of the first rotor is  $650$  kgm<sup>2</sup> where as of the second rotor is  $212$  kgm<sup>2</sup>. Determine the natural frequency of free torsional vibration of the system. The modulus of rigidity of shaft material may be taken as  $80$  GN/m<sup>2</sup>.

**UNIT IV**

- 17) The mass of each ball in a Wilson- Hartnell type of governor is  $2.5$  kg. The length of ball arm of each bell - crank lever is  $100$  mm whereas the length of the sleeve arm of bell - crank lever is  $80$  mm. The minimum equilibrium speed is  $200$  rpm when radius of rotation is  $100$  mm. When the sleeve is lifted to  $8$  mm, the equilibrium speed is  $212$  rpm. The stiffness of each of the springs connected to the balls is  $200$  N/m. The lever for the auxiliary spring is pivoted at the midpoint. Find the stiffness of auxiliary spring.

**[or]**

**18)** The mass of the turbine rotor of a ship is tonnes and has a radius of gyration of 0.60 m. The speed is 2000 r.p.m. The ship pitches  $6^\circ$  above and  $6^\circ$  below the horizontal position. A complete oscillation takes 30 seconds and the motion is simple harmonic.

Determine the following:

- a) Maximum gyroscopic couple
- b) Maximum angular acceleration of the ship during pitching, and
- c) The direction in which the bow will tend to turn when rising, if the rotation of the rotor is clockwise when looking from the left.

## UNIT V

**19)** A, B, C and D are four masses carried by a rotating shaft at radii 100, 125, 200 and 150 mm respectively. The planes in which the masses revolve are spaced 600 mm apart and the mass of B, C and D are 10 kg, 5 kg, and 4 kg respectively. Find the required mass A and the relative angular settings of the four masses so that the shaft shall be in complete balance.

**[or]**

**20)** A four cylinder vertical engine has cranks 150 mm long. The planes of rotation of the first, second and fourth cranks are 400 mm, 200 mm and 200 mm respectively from the third crank and their reciprocating masses are 50 kg, 60 kg and 50 kg respectively. Find the mass of the reciprocating parts for the third cylinder and the relative angular positions of the cranks in order that the engine may be in complete primary balance.

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**B.Tech. DEGREE EXAMINATION, APRIL/MAY 2015**  
**DYNAMICS OF MACHINERY**

**PART A - (10 x 2 = 20 marks)**

- 1) State the D'Alembert's principle.
- 2) State the function of flywheel in a prime mover.
- 3) What is meant by critical damping?
- 4) What is meant by free and forced vibrations?
- 5) Define the term: whirling speed of a shaft.
- 6) What is meant by torsionally equivalent shaft?
- 7) In what way do centrifugal governors differ from inertia governors?
- 8) What is the gyroscopic effect on a sea moving vessel?
- 9) What is meant by static balancing and dynamic balancing?
- 10) Why rotating masses are to be dynamically balanced?

**PART B - (5 x 11 = 55 marks)**

**UNIT I**

- 11) The following data relate to a horizontal reciprocating engine :
- Mass of the reciprocating parts = 120 kg  
Crank length = 90 mm  
Engine speed = 600 rpm  
Connecting rod mass = 90 kg  
length between centres = 450 mm  
Distance of Centre of mass from big end Centre = 180 mm  
Radius of gyration about an axis through Centre of mass = 150 mm  
Find the magnitude and direction of the inertia torque on the crankshaft when the crank has turned 30° from the inner – dead Centre.

**[or]**

- 12) A horizontal gas engine running at 210 rpm has a bore of 220 mm and a stroke of 440 mm, the connecting rod is 924 mm long and the reciprocating parts weigh 20 kg. When the crank has turned through an angle of 30° from the inner dead Centre, the gas pressures on the cover and the crank slides are 500 kN/m<sup>2</sup> and 60 kN/m<sup>2</sup> respectively. Diameter of the piston rod is 40 mm. determine
- a) Turning moment on the crank shaft
  - b) Thrust on the bearings

- c) Acceleration of the flywheel which has a mass of 8 kg and radius of gyration of 600 mm while the power of the engine is 22 kW.

**UNIT II**

- 13) Define the terms vibration isolation and transmissibility. Explain with the help of transmissibility Vs frequency ratio curves at various damping ratios.
- [or]**
- 14) A vibrating system consists of a mass of 50 kg, a spring with a stiffness of 30 kN/m and a damper. The damping provided is only 20% of the critical value. Determine the
- a) damping factor
  - b) critical damping coefficient
  - c) natural frequency of damped vibrations
  - d) logarithmic decrement
  - e) ratio of two consecutive amplitudes.

**UNIT III**

- 15) Describe the Dunkerley's method to find the natural frequency of a shaft carrying several loads.

**[or]**

- 16) A rotor has a mass of 12 kg and is mounted midway on a 24 mm diameter horizontal shaft supported at the ends by two bearings. The bearings are 1 m apart. The shaft rotates at 2400 rpm. If the Centre of mass of the rotor is 0.11 mm away from the geometric Centre of the rotor due to a certain manufacturing defect, find the amplitude of the steady-state vibration and the dynamic force transmitted to the bearing E = 200 GN/m<sup>2</sup>.

**UNIT IV**

- 17) Describe the simple function of a simple Watt governor. What are its limitations?

**[or]**

- 18) Explain in what way the gyroscopic couple effects affects the motion of an aircraft while taking a turn.

## UNIT V

- 19) The cylinder axes of a V-engine are at right angles to each other. The weight of each piston is 2kg and of each connecting rod is 2.8 kg. The weight of the rotating parts like crank webs and the crank pin is 1.8 kg. The connecting rod is 400mm long and its Centre of mass is 100mm from the crank pin Centre. The stroke of the piston is 160 mm. show that the engine can be balanced for the resolving and the primary Force by revolving counter mass, also, find the magnitude and the position if the Centre of mass from the crankshaft Centre is 100 mm. what is the value of the resultant secondary force if the speed is 840 rpm?

**[or]**

- 20) Determine the unbalanced forces and couples in case of following in-line engines:
- a) Two – cylinder engine.
  - b) Four – cylinder four stroke engine.

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